Using AI to assist high fidelity multiphase flow modeling in making atomization and spray predictions.

1. Background/Introduction

1.1 Relevance to DOE Genesis Mission

This proposal responds directly to "The Genesis Mission: Transforming Science and Energy with AI," specifically Topic 21, "Artificial Intelligence in Fluid Flow for Energy Components and Technologies," and Focus Area 21A, "Physics-Informed AI for complex fluid phenomena."

We seek support to develop AI-enhanced numerical methods for the simulation of atomization and spray processes that are central to combustion in aircraft engines [1] and rotating detonation engines used in rocket propulsion [2, 3]. These flows remain among the most challenging problems in computational fluid dynamics because they involve strongly coupled, multiscale interfacial dynamics that are difficult to resolve efficiently without sacrificing predictive fidelity.

Our approach is to place AI at the center of an efficient computational framework, using existing high-fidelity simulations and targeted experimental data to train sub-scale models that can generalize across a broader range of atomization and spray problems, including changes in geometry, fluid properties, and operating conditions. In direct alignment with Topic 21A, we will: (1) develop the capability to assimilate data from high-fidelity simulations and/or experiments into our existing multiphase-flow methodology [1–9]; and (2) use AI to learn dynamic and non-local sub-scale corrections that predict the gap between under-resolved simulations and super-resolved high-fidelity simulations and/or experiments.

The overarching goal is to create AI-generated sub-scale models that reduce the frequency with which computationally intensive high-fidelity simulations must be used, while preserving physical consistency and predictive capability. The broader impact of this effort extends well beyond the immediate target problems. In energy applications, accelerated multiphase simulation can improve predictive capability for fuel injection, combustion, boiling, evaporation, condensation, and advanced propulsion systems. More broadly, the proposed integration of physics-based simulation with data-driven learning supports the development of **adaptive digital environments** in which models, data, and computation are tightly coupled to accelerate scientific discovery and energy innovation.

1.2 Proposed Research in the context of the current state-of-the-art

Multiphase flows with *sharp, dynamically evolving interfaces* play a central role in energy conversion and transport systems, including fuel injection, spray combustion, rotating detonation engines, and phase-change-driven thermal processes [1–5, 10]. These flows are governed by *tightly coupled nonlinear interactions* among turbulence, interfacial forces, thermodynamics, and material response across a wide range of spatial and temporal scales, and are critical for identifying distinct mass and momentum processes at evolving interfaces. Their accurate prediction remains **one of the most challenging problems in computational science**.

Over the past several decades, semi-implicit **sharp-interface methods** have matured into a rigorous and reliable framework for modeling such systems. Foundational work by [11] established level set methods for tracking evolving interfaces in multiphase flow. Later advances produced conservative formulations [12] that significantly improved *mass conservation and robustness*, especially in the presence of topology change.

A major advance in this area is the **moment-of-fluid (MOF) [13–15]** and cell-integrated semi-Lagrangian **compressible MOF framework developed by Jemison, Sussman, and Arienti [16]**. The compressible multiphase formulation developed by Jemison et al. introduced a unified method that conserves mass, momentum, and energy while capturing shocks and material discontinuities without Riemann solvers. The material interfaces in [16] are represented sharply while preserving volume to machine precision in zero Mach number regions. Our approach enables simulations across compressible and incompressible fluids through asymptotic-preserving semi-implicit pressure updates. Further developments demonstrated accurate treatment of *multi-material junctions* and surface tension effects in complex geometries [2, 3, 5–8, 17]. However, the cost of resolving these multiscale interfacial processes at the fidelity required for predictive simulation remains high, making AI-enhanced modeling a natural and increasingly necessary next step.

This challenge becomes even more apparent in high-fidelity studies of fuel injection and atomization using DNS and LES approaches, which expose the *extreme computational cost* of resolving interface breakup, turbulence interaction, compressibility effects in realistic engine conditions, and, more importantly, sharp mass transfer at fluid interfaces. Such cost severely limits the utility of these methods for engineering design, where rapid turnaround and repeated evaluation over large parameter spaces are required. For problems such as liquid-jet-in-crossflow (JICF) atomization, an innovative computational approach, such as adaptive mesh refinement, provides only limited gains in practical multiphase simulations [10]. The strong feedback between chaotic gas-phase motion and liquid breakup makes this class of problems difficult to accelerate solely through refinement, motivating the need for new AI-enabled modeling and solution strategies. This motivates a new vision for AI-enabled solution strategies that preserve the unique predictive strengths of these methods while overcoming fundamental computational bottlenecks.

One possible avenue of improvement we considered, as well as others [7, 18], was to replace costly geometric operations in multiphase flow algorithms with surrogate AI-generated models. The key principle is that surrogate models must not compromise on the fidelity and stability of the numerical algorithm. In particular, for volume tracking, or moment of fluid methods, it is essential that the "linearity preserving" and "volume preserving" properties of a multiphase flow system not be violated [13, 14, 19–21]. Yet, what was discovered by us [7], and also mentioned in [18] is that these AI generated surrogate algorithms, at best, provide a marginal benefit. We also bring attention to the following articles which give evidence that surrogate AI models will never beat finely tuned optimized algorithms [22–25].

Another possible direction is to pursue meshless methods such as PINN [26, 27] or SPH [28, 29]. However, these approaches, and pure meshless methods more generally, do not inherently preserve volume. This is a fundamental limitation, because in the low-Mach-number limit the fluid volume should remain conserved. Notably, the volume error reported for PINN in [26] was *worse* than that of classical level set methods. In this sense, they revisit the main criticism of early level set methods: lack of strict volume conservation.

It is proposed herein to make two new developments: (I) we shall improve our existing compressible multiphase flow algorithm [2, 3, 16] by developing an accompanying volume and mass preserving compressible multiphase flow nudging capability for the assimilation of multiphase flow data from either high-resolution, high-fidelity simulations, and/or experiments [30]. (II) We shall use the nudging correction generated from (I) to train a neural network to derive sub-scale models appropriate for underresolved multiphase flows. It is expected that our sub-scale model will be general for a wide range of multiphase flows.

It is proposed herein to pursue two tightly connected advances: (I) we shall augment our existing compressible multiphase flow algorithm [2, 3, 16] with a volume- and mass-preserving compressible multiphase-flow nudging capability for the assimilation of data from high-resolution, high-fidelity simulations and/or experiments [30]. (II) We shall then use the nudging correction generated in (I) to train a conditioned AI model to construct sub-scale models for underresolved multiphase flows. The resulting sub-scale model is expected to generalize across a broad class of multiphase flow problems.

2. Project Objectives

The project objectives are twofold and directly aligned with the proposed AI-enabled multiphase-flow framework.

. 1: We will advance our existing compressible multiphase flow algorithm [2, 3, 16] by developing a compressible multiphase-flow nudging capability for assimilating data from high-resolution, high-fidelity simulations and/or experiments. The proposed nudging algorithm will enable volume- and mass-preserving "interface nudging" tailored to the complex atomization and spray problems considered here. Specifically, we will implement the framework developed by Dommermuth et al. [30], in which the nudging term takes the form $-\frac{p_s}{\rho}\nabla H$, where p_s is a nudging stress acting normal to the interface and is proportional to the difference between the simulated level set function and an assimilated level set function. (II) We will then use the nudging correction generated in (I) to train a neural network to derive sub-scale models for underresolved multiphase flows. The resulting model is expected to generalize across a broad range of multiphase flow regimes.

Our targeted canonical problems are: (a) atomization of a fuel jet in crossflow (JICF), which is directly relevant to combustion in aircraft engines [1]; and (b) the shock cylinder/drop problem, which is relevant to rotating detonation engines used in rocket propulsion [2, 3].

The project objectives are to (I) improve our existing compressible multiphase flow algorithm [2,3,16] by developing an accompanying compressible multiphase flow nudging capability for the assimilation of multiphase flow data from either high resolution, high fidelity simulations, and/or experiments. The nudging algorithm we propose to develop will enable volume and mass preserving "interface nudging" suitable for the complex atomization and spray problems that we consider. We propose to implement the algorithm developed Dommermuth et al [30]. In (31) of [30], the nudging term is $-\frac{p_s}{\rho}\nabla H$ in which p_s is a nudging stress that acts normal to the interface. p_s is proportional to the difference between the simulation level set function and an assimilated level set function. (II) We shall use the nudging correction, generated from (I), in order to train a neural network in order to derive sub-scale models appropriate for underresolved multiphase flows. It is expected that our sub-scale model will be general for a wide range of multiphase flows.

Our targeted canonical problems are (a) the atomization of a fuel Jet in a Cross Flow problem (JICF) which address combustion in aircraft engines [1] and (b) the shock cylinder/drop problem which address rotating detonation engines used by rockets [2, 3].

3. Proposed Research and Methods; Physics-Constrained Data Driven Model

The proposed research builds directly on prior work by PIs **Sussman, Shoele and Arienti** [**5–8, 16**], particularly the *cell-integrated semi-Lagrangian framework* and *moment-of-fluid interface representation*, which enable accurate tracking of deforming interfaces and efficient handling of *compressibility and stiffness*. Materials are represented using Volume fractions (zero order moment) and centroids (first order moment). Interfaces are captured. The governing equations include conservation of mass, momentum, energy, and species, with extensions for viscoelasticity, elasto-plasticity, and phase change. Interface conditions enforce *stress balance, velocity discontinuities due to mass transfer, and thermodynamic equilibrium.* Surface tension effects are modeled through *curvature-dependent forces*, and phase change is driven by *temperature and concentration gradients and corresponding nonlinear surface mass and chemical transport equations.*

In the proposed physics-constrained, solver-embedded learning framework, (1) high fidelity computational data and/or experimental data are assimilated via nudging into a coarse-grid simulation of two-phase flows, and (2) unresolved turbulence and interfacial dynamics are modeled using neural networks trained on the nudging source terms. Unlike conventional Physics-Informed Neural Network approaches that attempt to directly approximate the solution field, the present method retains the full structure of a conservative MOF-based two-phase flow solver, ensuring sharp interface representation, strict mass conservation, and compatibility with established numerical methods. The neural network is instead tasked with learning closure interfacial and bulk force corrections which will be embedded directly into the governing equations at each time step. Crucially, additional interface-aware constraints ensure that corrections remain localized and consistent with geometric quantities such as interface normals and curvature, addressing a major limitation of existing data-driven models in multiphase flows. When high-fidelity data are available, a compressible multiphase data assimilation nudging technique is used to align the coarse solution with reference dynamics, enabling accurate reconstruction of key flow features at a fraction of the computational cost. We note that when material interfaces are nudged, we ensure volume and mass preservation using the algorithms developed by Dommermuth et al [30] (see their equation (31)).

In this work, since independently evolved coarse-grid and high-fidelity two-phase simulations can rapidly diverge due to nonlinear amplification of interfacial errors, the raw mismatch between them does not provide a clean learning target for unresolved physics. Instead, the coarse solution is periodically assimilated toward the restricted high-fidelity state, $Q^{n+1}_{hf\to c}$, and the resulting aligned coarse state is used as the initial condition for one-step coarse-grid advancement. The discrepancy between this uncorrected coarse update and the corresponding high-fidelity evolution defines the target correction for the neural network. This strategy ensures that the learned model captures missing subgrid and interfacial physics, rather than compensating for accumulated trajectory drift.

The neural network is trained to infer unresolved interfacial and near-interface corrections from coarse-grid flow fields and reconstructed interface features, enabling the coarse solver to recover high-fidelity dynamics that are otherwise lost at reduced resolution.

Let Q^n_a denote the assimilated coarse state at time t_n, and let $Q^{n+1}_{c,unc} = G_{\mathrm{MOF}}(Q^n_a)$ be the corresponding uncorrected one-step coarse update. The target correction is defined as the one-step discrepancy between the restricted high-fidelity evolution and the uncorrected coarse prediction, $F^n_{\mathrm{corr}} = (Q^{n+1}_{hf\to c} - Q^{n+1}_{c,unc})/\Delta t$. A neural network $\mathcal{N}_\theta$ is then trained to approximate the mapping from coarse-grid flow and interface features $z^n = \Phi(Q^n_a, \Gamma^n_a)$ to this correction, where Γ^n_a is obtained from the MOF reconstruction of the assimilated moments. The training objective minimizes the discrepancy between predicted and target corrections, $\mathcal{L}(\theta) = \sum_n \|W \cdot (\mathcal{N}_\theta(z^n) - F^n_{\text{corr}})\|^2 + \mathcal{R}(\theta)$, incorporating optional weighting W and regularization $\mathcal{R}(\theta)$ concentrated near the interface.

Because the dominant modeling errors in coarse-grid two-phase simulations occur in mixed cells and in a narrow band surrounding the reconstructed interface, both supervision and regularization of the neural network are concentrated in these regions during training. Specifically, (1) the supervised loss is weighted by an interface-aware indicator so that discrepancies in interfacial regions contribute more strongly than errors in the bulk phases, where the coarse solver is typically already accurate. This weighting can be implemented using either a binary interface mask or a smooth distance-based function that increases the contribution of cells closer to the interface. (2) Physically motivated regularization terms are introduced to constrain the structure of the learned correction: 2a: A localization penalty is applied to discourage the network from producing spurious corrections away from the interface. 2b: A smoothness regularization is imposed within the interface band to prevent nonphysical oscillations and enforce spatial correction along the interface. 2b: A magnitude constraint is included to bound the correction and avoid overfitting or instability.

For interfacial force corrections, the third regularization term (3) is an additional geometric regularization that enforces alignment with the reconstructed interface normal, ensuring consistency with the underlying physics of interfacial interactions.

Together, these interface-focused weighting and regularization strategies ensure that the neural network learns physically meaningful, localized corrections associated with unresolved interfacial and subgrid-scale processes, while avoiding contamination from bulk-region noise or unphysical behavior.

4. Milestones in the Nine Months

1. (Sussman Months 1-3): Enhance the existing compressible MOF algorithm [2, 3, 5, 16] by adding the ability to assimilate, via nudging, compressible multiphase interface, velocity, and temperature data. Volume and mass conservation is ensured during the nudging procedure via the algorithm reported in earlier work by Dommermuthg et al [30].

2. (Arienti Months 1-3): Develop routines that import high-resolution data [2, 3] into actively running coarse grid simulations. Write restriction routines which carry out the operation: $Q^{n+1}_{hf\to c}$.

3. (Shoele Months 1-3): Develop the neural network algorithm which minimizes the discrepancy between predicted and target corrections, $\mathcal{L}(\theta) = \sum_n \|W \cdot (\mathcal{N}_\theta(z^n) - F^n_{\text{corr}})\|^2 + \mathcal{R}(\theta)$, incorporating optional weighting W and regularization $\mathcal{R}(\theta)$ concentrated near material interfaces.

4. (Arienti, Sussman, Shoele Months 4-6): Integrate the developments in items 1-3 above: nudging data assimilation, AMReX data base management, AI generated sub-scale models. Diagnostic testing on 2D atomization and spray from shock-cylinder interaction. Verification and validation.

5. (Arienti, Sussman, Shoele Months 7-9): Same as task 4, except fully 3D problems will be investigated)

5. Data Sources and Models

The current version of the computer code that is used to generate results shown in Figure 3 of [1], Figures 4 and 6 of [3] and Figures 5 and 11a of [2] is stored on Git Hub:
`https://github.com/msussman42/amrex_implicit_interfaces.git`
The data in Figure 3 of [1], and Figure 4 of [3] can be regenerated from scratch within 30 days. The remainder of the data is stored by Marco Arienti on national Lab computers.

6. Decision Gate Metrics

We shall train our neural network subscale models on the 2D/3D water and Galinstan shock-cylinder interaction problems (Figures 4,6,7 of [3]). The effective fine grid resolution for the “high fidelity” simulation is $3072 \times 2304 \times 768$ (3D). The effective fine grid resolution for the “low fidelity” simulation is $384 \times 288 \times 96$. After training, we shall rerun the water and galinstan cases, except using the trained models instead of assimilating the “high fidelity” data. Then we shall compare the respective interfaces between (a) coarse grid, assimilated results, (b) coarse grid, model results, and (c) high fidelity results. Finally, we shall create two new cases, varying the Weber number and Reynolds number. For each of the new cases we shall compare to the high fidelity model: (a) coarse grid, no model, (b) coarse grid with model. The interface errors are computed using the symmetric difference error [15].

Appendix 1: Bibliography and References Cited
phase I
Using AI to assist high fidelity multiphase flow modeling in making atomization and spray predictions.
PI: Mark Sussman, Department of Mathematics, Florida State University
CO-PI: Kourosh Shoele, FAMU/FSU Mechanical Engineering
CO-PI: Marco Arienti, Sandia National Labs, Livermore

References

[1] M Arienti, X Li, MC Soteriou, CA Eckett, M Sussman, and RJ Jensen. Coupled level-set/volume-of-fluid method for simulation of injector atomization. *Journal of propulsion and power*, 29(1):147–157, 2013.

[2] Marco Arienti, Everett A Wenzel, Daniel R Guildenbecher, and Steven J Beresh. Simulation and analysis of droplet aerobreakup in a ballistic wave system. *AIAA journal*, 63(8):3309–3320, 2025.

[3] Marco Arienti, Matthew Ballard, Mark Sussman, Yi Chen Mazumdar, Justin L Wagner, Paul A Farias, and Daniel R Guildenbecher. Comparison of simulation and experiments for multimode aerodynamic breakup of a liquid metal column in a shock-induced cross-flow. *Physics of Fluids*, 31(8), 2019.

[4] Marco Arienti and M Sussman. An embedded level set method for sharp-interface multiphase simulations of diesel injectors. *International Journal of Multiphase Flow*, 59:1–14, 2014.

[5] Everett A. Wenzel and Marco Arienti. A conservative, interface-resolved, compressible framework for the modeling and simulation of liquid/gas phase change. *Journal of Computational Physics*, 477:111957, 2023.

[6] Gautam Maurya, Yang Liu, Mark Sussman, and Kourosh Shoele. Drop transmission after the impact on woven fabrics. *International Journal of Multiphase Flow*, 179:104909, 2024.

[7] Zhouteng Ye, Cody Estebe, Yang Liu, Mehdi Vahab, Zeyu Huang, Mark Sussman, Alireza Moradikazerouni, Kourosh Shoele, Yongsheng Lian, Mitsuhiro Ohta, et al. An improved coupled level set and continuous moment-of-fluid method for simulating multiphase flows with phase change. *Communications on Applied Mathematics and Computation*, 6(2):1034–1069, 2024.

[8] Mehdi Vahab, Mark Sussman, and Kourosh Shoele. Fluid-structure interaction of thin flexible bodies in multi-material multi-phase systems. *Journal of Computational Physics*, 429:110008, 2021.

[9] Chaoxu Pei, Mehdi Vahab, Mark Sussman, and M Yousuff Hussaini. A hierarchical space-time spectral element and moment-of-fluid method for improved capturing of vortical structures in incompressible multi-phase/multi-material flows. *Journal of Scientific Computing*, 81(3):1527–1566, 2019.

[10] Xiaoyi Li and Marios C Soteriou. High fidelity simulation and analysis of liquid jet atomization in a gaseous crossflow at intermediate weber numbers. *Physics of Fluids*, 28(8), 2016.

[11] Mark Sussman, Peter Smereka, and Stanley Osher. A level set approach for computing solutions to incompressible two-phase flow. *Journal of Computational physics*, 114(1):146–159, 1994.

[12] Mark Sussman and Elbridge Gerry Puckett. A coupled level set and volume-of-fluid method for computing 3d and axisymmetric incompressible two-phase flows. *Journal of Computational Physics*, 162(2):301–337, 2000.

[13] Vadim Dyadechko and Mikhail Shashkov. Moment-of-fluid interface reconstruction. *Los Alamos report LA-UR-05-7571*, 2005.

[14] Hyung Taek Ahn and Mikhail Shashkov. Adaptive moment-of-fluid method. *Journal of Computational Physics*, 228(8):2792–2821, 2009.

[15] Matthew Jemison, Mark Sussman, and Mikhail Shashkov. Filament capturing with the multimaterial moment-of-fluid method. *Journal of Computational Physics*, 285:149–172, 2015.

[16] Matthew Jemison, Mark Sussman, and Marco Arienti. Compressible, multiphase semi-implicit method with moment of fluid interface representation. *Journal of Computational Physics*, 279:182–217, 2014.

[17] Guibo Li, Yongsheng Lian, Yisen Guo, Matthew Jemison, Mark Sussman, Trevor Helms, and Marco Arienti. Incompressible multiphase flow and encapsulation simulations using the moment-of-fluid method. *International Journal for Numerical Methods in Fluids*, 79(9):456–490, 2015.

[18] Aaron Mak and Mehdi Raessi. A machine learning approach to volume tracking in multiphase flow simulations. *Fluids*, 10(2), 2025.

[19] Elbridge Puckett. A volume-of-fluid interface reconstruction algorithm that is second-order accurate in the max norm. *Communications in Applied Mathematics and Computational Science*, 5(2):199–220, 2010.

[20] Elbridge Puckett. On the second-order accuracy of volume-of-fluid interface reconstruction algorithms: convergence in the max norm. *Communications in Applied Mathematics and Computational Science*, 5(1):99–148, 2010.

[21] Elbridge Puckett. Second-order accuracy of volume-of-fluid interface reconstruction algorithms, ii: An improved constraint on the cell size. *Communications in Applied Mathematics and Computational Science*, 8(1):123–158, 2014.

[22] Thomas Milcent and Antoine Lemoine. Moment-of-fluid analytic reconstruction on 3d rectangular hexahedrons. *Journal of Computational Physics*, 409:109346, 2020.

[23] Steven Diot and Marianne M. François. An interface reconstruction method based on an analytical formula for 3d arbitrary convex cells. *Journal of Computational Physics*, 305:63–74, 2016.

[24] Matthieu Ancellin, Bruno Després, and Stéphane Jaouen. Extension of generic two-component vof interface advection schemes to an arbitrary number of components. *Journal of Computational Physics*, 473:111721, 2023.

[25] Joaquín López and Julio Hernández. Analytical and geometrical tools for 3d volume of fluid methods in general grids. *Journal of Computational Physics*, 227(12):5939–5948, 2008.

[26] Mathieu Mullins, Hamza Kamil, Adil Fahsi, and Azzeddine Soulaïmani. Physics-informed neural networks for solving moving interface flow problems using the level set approach. *Physics of Fluids*, 37(10), 2025.

[27] Darioush Jalili, Seohee Jang, Mohammad Jadidi, Giovanni Giustini, Amir Keshmiri, and Yasser Mahmoudi. Physics-informed neural networks for heat transfer prediction in two-phase flows. *International Journal of Heat and Mass Transfer*, 221:125089, 2024.

[28] M.Z. Wang, Y. Pan, X.K. Shi, J.L. Wu, and P.N. Sun. Comparative study on volume conservation among various sph models for flows of different levels of violence. *Coastal Engineering*, 191:104521, 2024.

[29] Prapanch Nair and Gaurav Tomar. Volume conservation issues in incompressible smoothed particle hydrodynamics. *Journal of Computational Physics*, 297:689–699, 2015.

[30] Douglas G Dommermuth, Christopher D Lewis, Vu H Tran, and Miguel A Valenciano. Direct simulations of wind-driven breaking ocean waves with data assimilation. *arXiv preprint arXiv:1410.0418*, 2014. 30th symposium on naval hydrodynamics, Hobart Australia, November 2014.

Appendix 2: Facilities and Other Resources
phase I
Using AI to assist high fidelity multiphase flow modeling in making atomization and spray predictions.
PI: Mark Sussman, Department of Mathematics, Florida State University
CO-PI: Kourosh Shoele, FAMU/FSU Mechanical Engineering
CO-PI: Marco Arienti, Sandia National Labs, Livermore

Laboratory Resources: The FAMU-FSU College of Engineering performance site facilities used in this project are located within the innovation park at the joint college of engineering in Tallahassee. CO-PI Shoele is a member of Florida Center for Advanced Aero-Propulsion (FCAAP) and has a 270 square foot lab equipped with computers. Office space for graduate students are provided by FCAAP.

Computer (Shoele): CO-PI Shoele's laboratory is equipped with over a dozen PC workstations. Key software includes Tecplot, GridGen, Intel compilers. The department of Mechanical Engineering and FCAAP provide the software as well as printing facilities.

Computer (Sussman): PI Sussman has access in the math department to 8 4-core computers, 1 32 core machine, and 1 8-core machine. The 8-core machine has an Nvidea GPU card in it.

Computing Facilities: There are a number of major computing facilities at FSU-FAMU that are available to the PIs Sussman, Shoele, and students in FSU-FAMU for this project. 1) HPC Cluster: 647 node, 10772 core, cluster with CUDA GPU capability attached to 256 TB Panasas file system that is mounted on all of the HPC compute nodes over a dedicated 1Gbps network. 2) Spear cluster: 88 x86 64-bit process cores linked together by a QDR Infiniband network. Eight of the nodes also connect to a PCI chassis equipped with 16 NVIDIA Tesla gpGPU cards (m2050) for a total 7,168 GPU cores for a peak system performance of 19.8 TFLOPS. 3) CAPS Real Time Digital Simulator (RTDS): A parallel-processing machine (performance at over 110 GigaFLOPS) with "14- rack" system consisting of 380 parallel processors and employing optimized EMTP-based nodal network simulation methods.

Appendix 7: Letter of Commitment from Sandia Labs
phase I
Using AI to assist high fidelity multiphase flow modeling in making atomization and spray predictions.
PI: Mark Sussman, Department of Mathematics, Florida State University
CO-PI: Kourosh Shoele, FAMU/FSU Mechanical Engineering
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Exceptional service in the national interest

Camron Proctor
Manager, Thermal Fluids Modeling and Simulation

April 14, 2026

To Whom it May Concern:

Sandia National Laboratories (Sandia) is pleased to participate with Prof. Mark Sussman (Department of Mathematics at Florida State University) in their response to DOE's Request for Applications "The Genesis Mission: Transforming Science and Energy with AI" (DE-FOA-0003612) for topic 21, "Artificial Intelligence in Fluid Flow for Energy Components and Technologies." The title of the proposal is "Using AI to assist high fidelity multiphase flow modeling in making atomization and spray predictions."

Mark Sussman has requested Sandia's unique capabilities to be included in their response to this Request for Applications. Subject to available funding and resources, Sandia anticipates participating in mutually agreed tasks, which will be performed by staff in the Thermal/Fluids Science and Engineering Department. The objective will be to develop AI-enhanced sharp-interface numerical methods that can reduce the computational cost in R&D of advanced combustion devices (such as jet engines and rotating detonation rockets). This activity will take place using existing DOE-developed simulation codes. Sandia staff will provide critical domain knowledge to anchor the choice of the user cases, while also assisting in the setup and postprocessing of the simulations. Crucially, the staff will test the AI-augmented workflow in comparison with existing best practices.

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This information is provided in accordance with applicable DOE policy regarding Sandia's participation in federal government announcements. The Sandia point of contact for this potential collaboration is Marco Arienti, Sandia National Laboratories, P.O. Box 969, Livermore CA 94551.

Sincerely,
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project summary
phase I
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CO-PI: Marco Arienti, Sandia National Labs, Livermore

The atomization and evaporation of fuel sprays is a fundamental process that takes place in internal combustion engines. This proposal focuses on new numerical methods for carrying out "digital twin" atomization experiments, leading to the improvement in the efficiency and/or performance of jet engines and rotating detonation rocket engines.

Effective Numerical methods and experiments alike must reliably measure multiscale processes ranging from millimeters to nanometers. Macro scale sprays break up into sheets and ligaments, which in turn break up into a mist. Evaporation turns the mist into a gaseous fuel which is ready to be reacted with oxidizer. The spray breakup process, along with convective heat transfer from evaporation and preceding reactions, induces a feedback mechanism in which the resulting turbulent flow field affects the breakup. At the same time, for rotating detonation engines, not all of the droplets need to be atomized prior to shock wave interaction. Remaining micro-scale sized droplets that survived the "pre-shock" atomization process are shattered by the supersonic detonation wave into a sub-micron "fog" almost instantly. At which point reactions occur, thereby generating the detonation wave for the next cycle. It is noted that in an optimized rotating detonation engine, the "pre-shock" remaining droplets must not be too large (so as to survive the shock wave interaction) and not too small either (so as not to have used too much energy in the pre-shock atomization phase).

It is with this backdrop that the reality is that both high-fidelity atomization and spray simulations and physical experiments are very expensive. This proposal's objective is to develop novel Artificial Intelligence (AI) methods for making the best use of data that is available. The proposed work consists of 3 parts: (1) data assimilation, (2) AI learning of multiphase flow sub-scale models, and (3) verification and validation.

1. Recent multi-phase flow data assimilation algorithms have been developed by Dommermuth et. al. for assimilating measured sea surface data for the purpose of improving incompressible multiphase flow predictions as well as enabling the discovery of hidden experimental states. We shall develop analogous data assimilation techniques with the extended capability for compressible multiphase flows.

2. The discrepancy between a coarse grid simulation, with data assimilation, and the corresponding high fidelity data, is used as training data for a neural network model in order to learn the coarse grid correction term, using solely coarse grid simulation data.

3. Coarse grid simulations are carried out for atomization and spray cases not included in the original data assimilation process (step 2). The coarse grid simulations are carried out with/without the aid of the neural network model developed in case 2 above.